HERIOT-WATT UNIVERSITY

F79MB Statistical Models B **Assessment Project 1 – 2024**

Your report for this project should be submitted through Turnitin by **3:30pm (local time at your campus) on Monday 26th February 2024**. All required data sets and a link to the submission page are available through "Assessed Project 1" in Modules on the course Canvas page. Please use the submission link provided to submit your work.

This project will account for **40%** of your grade for the course.

The total length of your report should be no more than 8 pages (11-point font, A4 size), including graphs and tables but excluding any appendices. Project title and student identification information should be in page 1, not a separate title page, and counts towards the 8-page limit. Please note that the **overall exposition/presentation is worth 8 marks** for this project.

You may consult with your lecturer or coursemates, but your report must be your own work. **Plagiarism** is a serious academic offence and carries a range of penalties, some very serious – see Appendix A. You are required to submit a Standard Declaration of Authorship form. The form is available through 'Assessed Project 1' on the course Canvas page. **You must sign this Declaration of Authorship for Project 1 before you can have access to the project 1 questions**.

- Answer all tasks in this project, making sure that you add appropriate context and that your answers have a clear and logical structure and are well presented.
- You should explain carefully your work for each task so that your work demonstrates understanding of the methodology and computations that you use.
- You should include clearly labelled and correctly referenced graphs and add appropriate comments.
- You should use `R` to perform the required analyses and produce suitable graphs. Unless otherwise stated you do not need to explain the `R` commands that you use. However you must put the `R` code in appropriate appendices at the end of your report, and should include appropriate commenting within your `R` code.
- See Appendix B for the marks allocated to overall exposition/presentation.
- **Late project submissions will be penalised according to the University Policy on Submission of Coursework. That is, work submitted after the deadline but within 5 working days will be subject to a 30% deduction from the mark awarded; work submitted more than 5 working days after the deadline will be awarded a mark of zero. No individual extensions are permitted. In the case where a student submits coursework up to five working days late and has valid mitigating circumstances, the mitigating circumstances policy will apply.**

You can expect to receive feedback on your work by 18th March 2024.

Tasks:

1. The data set in the `travelexpenses.txt` file, available on Canvas under `Modules > Assessed Project 1`, comprises work travel expenses (in US Dollars) for 166 employees of an IT company in Washington from years 2020 to 2023.

   (a) Present a histogram and appropriate numerical summary statistics for the data. Share insights on the data distribution, ensuring to include more than one comment.

   [2 marks]

   (b) Use QQ plots to explore whether the original data, the square root transformed data, or the (natural) log-transformed data, can be reasonably modelled using a Normal distribution.

   [4 marks]

   (c) For the square root transformed data, perform a chi-squared goodness-of-fit test to formally assess whether the transformed data can be reasonably modelled using a Normal distribution. For this test, divide the data into 7 cells and ensure equal probability across all intervals. Comment on your results and discuss whether they align with the QQ plot observed in 1(b).

   [4 marks]

   (d) Based on the QQ plots and the chi-squared goodness-of-fit test results in 1(b) and 1(c), identify any potential outliers in the square root transformed data, if present. Justify your answer.

   [1 mark]

   Then, discuss the implications of outliers (in general) on the analysis, suggest possible reasons for their occurrence, and propose methods for handling these outliers in the analysis.

   [3 marks]

   (e) Assuming the square root transformed data can be reasonably modelled by a Normal distribution, find a 95% confidence interval for the population median of the square root travel expenses using asymptotic theory.

   Next, obtain a 95% confidence interval for the population mean of the square root travel expenses based on asymptotic theory. Discuss your observations regarding these confidence intervals and provide justifications for selecting your preferred confidence interval.

   [4 marks]

   (f) Based on the analysis above, summarise your overall conclusions about the travel expenses data.

   [2 marks]

2. The data set in the `usedcar_price.txt` file, available on Canvas under `Modules > Assessed Project 1`, consists of the selling prices of BMW used cars (in £) in the United Kingdom of a random sample of 28 sales for year 2023.

   (a) Present appropriate numerical and graphical summaries, and comment on the distribution of the data.

   [2 marks]

   (b) Use non-parametric bootstrap methodology to obtain the empirical sampling distribution of the sample standard deviation of the prices of BMW used cars and present the distribution graphically. Comment on the results.

   Compute a non-parametric bootstrap 99% confidence interval for the population standard deviation of the BMW used car prices. Next, compute the corresponding confidence interval using asymptotic theory.

   [3 marks]

   (c) Discuss and comment on the validity of the confidence intervals for the standard deviation of the BMW used car prices obtained in 2(b).

   [2 marks]

   (d) An analyst from Arnold Clark car retailer claims that the mean selling price of a BMW used car in the UK for the year 2023 is £15,000 or less. To avoid making assumptions about the distribution of the data, perform a non-parametric bootstrap test to assess the validity of the analyst's claim using the provided data set.

   [2 marks]

   (e) The analyst also wants to consider the use of a different approach for testing the claim in 2(d). Perform an appropriate parametric test, and comment on the conclusion compared with that obtained in 2(d). Also briefly discuss any possible issues regarding the validity of the two tests.

   [3 marks]

[Overall exposition/presentation: 8 marks]

[Project total: 40 marks]

[END OF PROJECT]

Appendix A: Plagiarism

- Coursework reports must be written in a student's own words and any code in their coursework must be their own code. If some text or code in the coursework has been taken from other sources, these sources must be properly referenced.
- Failure to reference work that has been obtained from other sources or to copy the words and/or code of another student is plagiarism and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course.
- Students must never give hard or soft (electronic) copies of their coursework reports or code to another student. Students must always refuse any request from another student for a copy of their report and/or code.
- Sharing a coursework report and/or code with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.
- See `https://www.hw.ac.uk/uk/students/studies/examinations/plagiarism.htm` for more details.

Appendix B: Rubric for marks allocated to overall exposition/presentation

The 8 marks available for the exposition/overall presentation of your report will be awarded according to the scale below.

2 marks	• Some answers lack of clarity and logical structure • Some analyses may not relevant • Statements of conclusions are present but may not be very clear • Some statistical calculations and methodology set out for the reader • Some tables and figures relevant and referenced
4 marks	• Answers sometimes clearly and logically structured • Focus on key points but may include some superfluous/irrelevant analyses • Statements of conclusions generally suitable for a non-statistician • Statistical calculations and methodology are generally presented okay for the reader • Tables and figures mostly well chosen, clear, and correctly referenced • Some sources are clearly and correctly referenced, with R code included
6 marks	• Answers are mostly clear and logically structured • Clear focus on key points, avoiding superfluous/irrelevant analyses • Statements of conclusions suitable (wherever possible) for a non-statistician • Statistical calculations and methodology set out clearly for the reader • Tables and figures well chosen, clear, correctly referenced and easy to interpret • Sources used clearly and correctly referenced • R code included with comments
8 marks	• As in the previous category, but the answers are very clear, strong, and logically structured. Additionally, the work demonstrates mathematical sophistication and insight, with evidence of original thought and reasoning. It also maintains a highly focused approach on key points, without superfluous or irrelevant analyses. R code is included with comments and without superfluous coding.